

Problem Set 3

POLS W4732

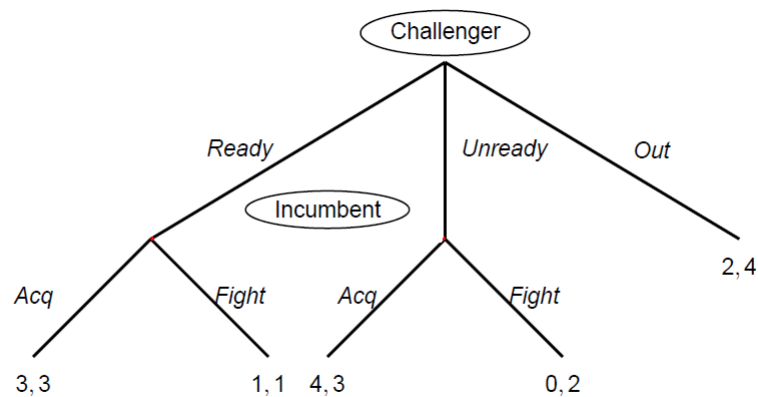
Fall 2026

Due October 16 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

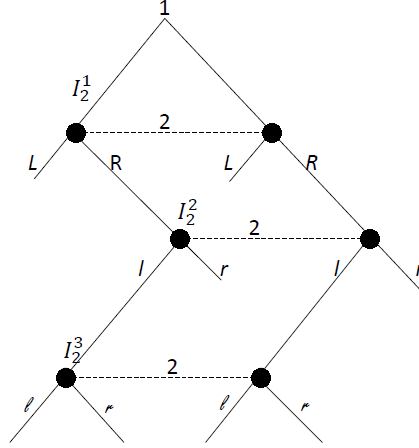
Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider the following version of the Entry Game



Using its strategic form representative, find the set of Nash Equilibria of this game.

Exercise 2. Consider the following game, where I_2^1, I_2^2, I_2^3 denote the information sets for Pl. 2. Let $s_2 \equiv (L, l, r)$ and $s'_2 \equiv (R, r, l)$, denote two pure strategies for Pl. 2. Find the behavioral strategy equivalent to the mixed strategy that puts equal probability (i.e., $1/2$ and $1/2$) on s_2 and s'_2



Exercise 3. Consider the Public Good Provision Game, under the assumption that both costs are drawn from a uniform distribution in $[\gamma - \frac{1}{2\psi}, \gamma + \frac{1}{2\psi}]$, with $\gamma - (2\psi)^{-1} < 1 < \gamma + (2\psi)^{-1}$

1. Characterize the unique Nash Equilibrium
2. How does the equilibrium threshold c^* change with ψ and γ ? Can you explain these results intuitively?
3. How does the equilibrium probability that the public good is provided change with ψ and γ ? Can you briefly explain these results intuitively?
4. Now suppose that instead of two players, the game features n players. For simplicity, also assume that $\psi = 1/2$ and $\gamma = 1$. How does the equilibrium threshold c^* change with n ? Can you briefly articulate an intuition for this result?
5. (**hard**) Under the same assumptions as in the previous point, how does the probability that the public good is provided change with n (provided n is arbitrarily large)? Can you briefly articulate an intuition for this result (even if you are unable to show this)?

Exercise 4. Consider the following simple model of war. There are two countries (1 and 2). Each has two actions: F (prepare for a fight) or N (accommodate). We define war as the outcome (F, F) . When both countries choose N , they obtain a payoff of 0 for sure. Choosing F incurs mobilization cost μ whether or not the war happens. If the war happens, the winner takes resources worth 1 from the loser. The winner is ex ante uncertain, and is the realization of a state of nature $\omega \in \{UR, DR, UL, DL\}$. When the state is either UR or DL , country 1 would win the war should it occur; otherwise, country 2 would win. We assume that each country only observes one of the two components of the state (e.g., when the state is UL country 1 receives signal “U”, country 2 receives signal “L”).

$$\begin{aligned}
 u_i(N, a_j, \omega) &= 0 \quad \forall (a_j, \omega) \in \{F, N\} \times \Omega \\
 u_i(F, N, \omega) &= -\mu \quad \forall \omega \in \Omega \\
 u_i(F, F, \omega) &\in \{1 - \mu, -1 - \mu\} \quad \forall \omega \in \Omega
 \end{aligned}$$

Notice that fighting is inefficient: $u_1(F, F, \omega) + u_2(F, F, \omega) = -2\mu$ for all $\omega \in \Omega$.

1. *Show that $(N(U), N(D), N(L), N(R))$ is a Bayesian equilibrium*
2. *Show that, if country 1 chooses F when its signal is U , it must be that the other country also chooses F with positive probability (either upon seeing R or upon seeing L , or both)*
3. *Show that if this is the case, at least one country has a profitable deviation*